

Prioritizing UI Items for Bidirectional Design System Development: A Suggested Set of Criteria and its Implementation

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ABSTRACT

Design systems provide detailed guidelines on UI items' appearance and behavior and serve as a primary reference for interface development. For bidirectional languages (e.g., Arabic and Hebrew), it is essential to highlight an additional parameter – the direction of the UI item. However, investigating the correct direction of all items could involve costly research. Hence, it is important to set priorities for such research. While prioritizing usability issues has long been attended by the HCI community, the specific problem of this research is new and has never been addressed. To deal with this challenge, we conducted five focus groups with HCI professionals and identified a set of prioritization criteria. We then used those criteria to prioritize the UI items for future empirical evaluation of their preferred directionality. In addition to these findings, we provide some insights from our experience of using focus groups for the prioritization task.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods.

KEYWORDS

BiDi, Bidirectional design, Design system, Focus Group, UI item direction, Usability Prioritization

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1 INTRODUCTION AND BACKGROUND

Recent years have seen design systems proliferation in technology companies around the world. By providing uniform shared design

and interaction standards about how UI items should look and behave, design systems guide designers and developers to improve consistency, interface usability, and design efficiency [3, 6, 26]. Yet, BiDirectional (BiDi) languages (e.g., Arabic, Hebrew, or Persian) pose an additional challenge to design systems. Those languages mix native right-to-left (RTL) script with left-to-right (LTR) elements (e.g., English words, numbers, or notations) [1, 4, 9]. Hence, to avoid user mistakes, inconsistency, and ambiguity, the BiDi design system should include an additional essential guideline – the recommended direction of the UI item. Should the item appear in an LTR or in RTL direction (see Figure 1)?

Due to linguistic differences between BiDi languages [16, 19], the intricate nature of BiDi users' perception and information processing [7, 8, 23, 24, 27], identifying the optimal direction of UI items becomes a daunting task, which in most cases requires usability study with native BiDi speakers [9]. The potential benefits of such research include, for example, meeting the needs of more than half a billion [5, 15, 25] BiDi language users. However, this, like all research, is subject to limited resources. Hence, there is a need to prioritize and cost-justify which UI items should be studied at greater urgency. Whereas cost-justification has been considered for usability practice and research [2], it has never been used towards studying the directionality of UI items in BiDi interactive systems.

In this paper, we report on a study we conducted to address this challenge. The purpose of the study was to prioritize research on UI items that emerged as potentially equivocal or misleading in the context of BiDi design. *Given the inevitably restricted research resources, which UI items should we study first to inform designers of BiDi design systems, applications, and websites? What are the criteria for these decisions?* To answer these meta-research questions, we conducted five focus groups with a total of 22 HCI professionals. The current paper describes the research procedure, data analysis, and findings. Our main contributions are: (1) a set of criteria to help decide the urgency of empirically evaluating the desired directionality of any of numerous UI items; (2) a prioritized list of possible UI items for the empirical evaluation.

2 METHOD

We conducted a series of five online focus groups (FG) with HCI professionals. Each FG included three to six participants. The FGs were based on semi-structured interviews, combined with individual expert judgment tasks, followed by a group discussion and resolution. We chose this method to let experts start by considering the issues and expressing their opinions independently [18]. A group discussion then followed to allow consideration of multiple viewpoints

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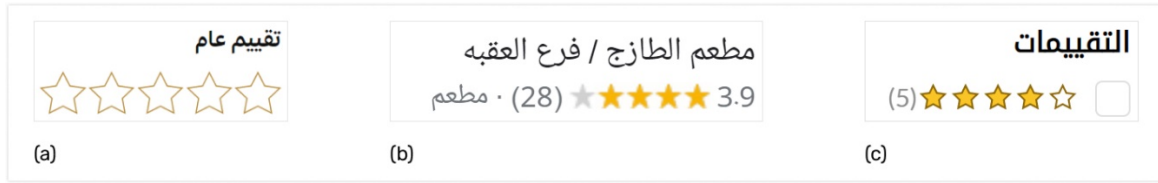


Figure 1: Examples of BiDi design complexity: (a) An ambiguous view of star rating from Amazon.com web site. The user can not know which side the rating starts without additional explanations. (b) RTL star rating from Google maps. (c) LTR star rating from iherb.com. Retrieved from Arabic version web sites on January 5, 2022.

to tease out new ideas and insights [17]. To better understand BiDi users, four of five FGs included experts who are native speakers of a BiDi language (Hebrew) [1, 2, 14]. We also had one focus group of experts (N=6) who are not familiar with BiDi languages to obtain a potentially different perspective on the research problem.

2.1 Procedure

The research procedure was approved by the institution’s Human Subject Research Committee. Before the FG, the participants completed an agreement form for participation and recording. The study was conducted from April to June 2021 using the Zoom platform for conversation and the Miro platform for individual tasks and group discussion. The participants used personal computers with web cameras. Two participants turned off their cameras due to the weak internet signal during the session. No participants left the discussion.

We developed a research space in Miro, with boards for individual work and group discussion (see an example of boards in supplementary materials). Each board included a list of 45 UI items (see App. and Suppl. materials), which was based on the list of challenging UI items in BiDi (Hebrew) websites [9]. The directionality of these UI items was used inconsistently (both right-to-left and left-to-right) in the most popular Hebrew interfaces, and for a major part of UI items, the optimal directionality definition was absent in the current BiDi design guidelines [9]. To facilitate navigation between the UI items, we divided them into six categories based on the classification suggested in Goldenberg & Tractinsky, 2021 [9]: quantity, order, and preference; text with numbers; time or flow; date (calendars); motion; space and navigation. We added six additional UI items to the research set to balance the number of items in each category. The additional items (forward icon, carousel, navigation drawer with “hamburger” icon, progress bar, process text, and process images flows) are inconsistently used in BiDi interfaces.

To ensure that the participants correctly understand the items, we have added illustrations for elements whose names are used differently in various design systems (e.g., for counter and toggle) and icons. We also illustrated the meaning of items belonging to the “text with numbers” category (e.g., “11.05-13.06” for a date range or “07:00-21:00” for a time range).

Each FG lasted between one and a half to two hours, was recorded and facilitated by a moderator (the first author). While signing up for the study participation, the participants were informed about the topic of the FG and its procedure, including individual tasks (first phase) and group discussions (second phase). We read out the

presented UI items and ensured that all participants understood what kind of UI item was used in the study.

In the individual task, we asked the participants to look at the personal board and define the directionality of which UI items should be verified in the study with BiDi users and further explained in the design systems and which UI items could be excluded from the research set (15 in total). The participants explained their decisions by writing comments on interactive sticky notes. During the individual task, participants could add other UI items, change categories names, and move UI items between groups (see Supplementary materials for the individual task instruction). Using Zoom’s facilities, the participants could communicate with the moderator for questions and clarifications both orally or through the chat. Participants unfamiliar with Miro (P14-P16) conducted a task in a Word document. After completing the individual task, the participants were invited to join a group decision on the same issue and vote on the inclusion of the UI item in the research set. The moderator managed the group board.

2.2 Participants

In total, 22 HCI professionals participated in the study: 16 UX designers, one UI designer, and five product managers. Sixteen participants resided and worked in a predominantly BiDi country (Israel) and had experience in BiDi (Hebrew and Arabic) interface design and usage. Six participants were Russian speakers who resided and worked in Russia (2) or Denmark (4) and did not have any experience in BiDi interface design and usage. The participants varied in professional experience, areas, and languages of designed projects (see Supplementary materials for the details), understood the UI items’ role, functionality, and usage specifics. All participants had high familiarity with Zoom and varied in the familiarity with Miro usage: three participants had never used the platform before, eight used the platform in their daily professional routine, other participants have used the platform several times before. The participants were able to terminate their participation in each study stage. Participation in the study was voluntary, not financially rewarded, and based on the participants’ deep interest in the research topic.

2.3 Analysis

The analysis was carried out by two researchers (the authors). The data was analyzed using the “Theming the data” method [22] on the first analysis cycle, Applied Thematic Analysis [11] on the second cycle, and then quantitative calculations [12]. The qualitative coding, categorization, and analysis were performed using Excel’s sorting, filtering functions and color highlighting functions [20, 22].

Table 1: Criteria for inclusion of UI items in the directionality evaluation research for BiDi Design System

#	Criterion	% Of all statements	Participants who used the criterion in argumentation
	High cost of errors	48%	All
	Design dependent UI item	23%	All, except P8
	Knowledge about UI item correct direction	20%	All participants, except P19-20
	Importance of the UI item	17%	All participants except P6
	Risk of inconsistent usage	12%	P1-P5; P7-P12; P14; P17-20
	Problematic UI item in general	7%	P1-P6; P9-15; P17-18
	Complex UI item	3%	P2-P6, P8, P12, P16

We prepared the data for analysis by recording all FG discussions data on a Word document. The quotes related to the research question (comments about the UI item selection in the individual boards and the statements during the group discussions) were further translated to English, transcribed, manually coded, and organized in a Microsoft Excel sheet. The descriptive information about the participants received a separate Excel Sheet.

The groups did not have to come to a unanimous decision about each UI item. In cases where the FG members could not agree about a specific UI item, we calculated the percentage of opinions “to include” and the percentage of “to remove” for each of the UI items (see the third column in the App.).

We observed that in most discussions, the experts grouped some UI items in their arguments. All participants grouped the *backward* and *forward* icons (buttons); the *undo* and *redo* icons (buttons); and the *arrival* and *departure* icons. More than 70% of the participants combined in their argumentations the *bus*, *delivery*, *phone*, and *shopping trolley* icons. Similarly, the participants tended to group range items – *date*, *time*, *size*, and *temperature*. Consequently, we analyzed the statements about the combined UI items according to their respective groups.

3 RESULTS AND DISCUSSION

3.1 Determining criteria for including UI items in the research

In response to the study’s main objective, seven criteria emerged from the analysis of the FG discussions. The criteria are based on the participants’ statements. They are presented in Table 1 and described below with relevant quotes from the participants’ statements.

1. **High cost of errors.** An incorrect direction of a UI item in a BiDi system will definitely result in an interaction error. Such errors can sometimes have critical consequences. Hence, the more consequential the incorrect use of an item, the more important it is to evaluate its correct direction and to standardize how it should be used. For example, the following issues were mentioned by the participants as having potentially critical consequences:

1. Information misunderstanding (e.g., P16 about the *send* icon: “Users may get confused and think that they are deleting the text instead of sending”).

2. Performing a wrong action that cannot be fixed (e.g., P4 about the *star rating* element: “You might rank a product incorrectly, in many cases there is no way to fix it”).
3. Missing information (e.g., P10, P17 about the *carousel* element: “If you swipe in the wrong direction, you might miss the content”).

In contrast, some issues due to the wrong direction of the UI item may not lead to critical consequences. Examples of such issues mentioned by the participants were:

1. A mistake that could be fixed (e.g., P13, P18 about the *paginator* item: “There is an indication – page numbers, so it is not so scary, you can eventually understand and perform the action correctly”).
2. User annoyance: “It’s annoying maybe, but it’s pretty clear what to do. There is a difference between being upset and not knowing what to do” (P4 about the *counter* item).
3. Quick error understanding (e.g., P3 about the *counter* item: “I clicked once, I’ve seen it does not work, and then I clicked another button. Not that dramatic”).

In some cases, the participants disagreed about the severity of having items incorrectly displayed. For example, the consequence “increasing time for task performance” was used both as supporting the importance of evaluating a UI item direction (e.g., P2 about *counter*: “It would take time to figure out if you counted incorrectly. it must be in accordance with what a regular user does”), as well as downplaying it: “Get along with incorrect calendar direction, it is true that it may be less convenient to read it, it may take a long time, but you get used to it” (P3 about a *calendar that does not include an indication of the weekdays as a clue for directionality*).

2. **Design-dependent UI item.** The criterion means that if the UI item’s direction could be misunderstood due to faulty design, it should be evaluated and included in the BiDi design system. A faulty design can stem from various reasons, such as a UI item without a label, values, or color indication (e.g., “If a *toggle* comes without a label, or its status is not colored well, it is very difficult to understand it correctly” (P6); small item size (“If the arrows are small it will be critical – at first glance, the user might misunderstand the flow” (P7, P18 about *process images flow*)).

This criterion is especially relevant for ambiguous UI items, which could be interpreted as both LTR and RTL: “I never know how to read the time range string: from right-to-left or from left-to-right. If you do not have any indicator, you will not be able to understand it correctly” (P13). According to the participants,

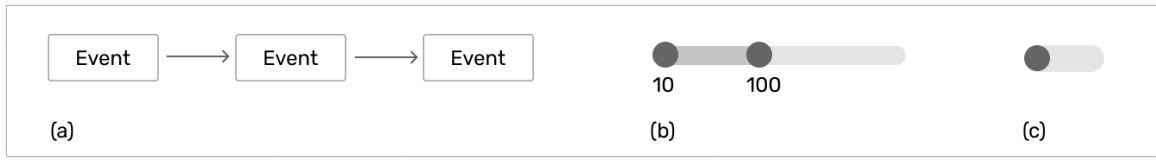


Figure 2: UI items found as most “Design dependent”: (a) Process text flow. (b) Double slider. (c) Toggle. All items appear in this example in their LTR version

among the ambiguous UI items were the following: “backward” and “forward,” “back” and “next,” “redo” and “undo” icons (buttons); date, time ranges; “Sale X+Y” strings [9]; progress bar; toggle; carousel.

Design-dependent UI items may also refer to the platform of usage. For example, the lack of standard about the place of the *close* button in the dialog window (top left or top right) coupled with poor design might hinder performance on the web, but it is not a critical issue in mobile. “On a mobile, it really does not matter to me, but on the desktop, if the Close button is not very large and prominent, it will take me a time to find where it is” (P5).

Interestingly, in all FGs we observed a process of realization that to be understood, UI items often require immaculate design. For example, in three FGs the discussions about the *double slider* started with the statement that this item has a clear visual indication that helps in understanding it: “The direction does not matter, it is usually shown with values, and therefore it is clear enough” (P6). Yet, the experts began to question what would happen if the *double slider* was not designed well enough: “But what if there are no values?” (P8). Consequently, the group decided to increase the priority of that item: “If there is no good visual indication, its directionally must be consistent across all interfaces” (P8).

Design dependence was the primary criterion for argumentation about the following UI items (see Fig. 2):

1. *Process text flow*. If the arrows are thin, small, pale, or inconspicuous, it is crucial to display process flow (with text) in the correct direction.
2. *Double slider*. The information processing in a *double slider* depends on whether the values and the labels are clearly shown and consistent. If they are not shown correctly, it is essential to display the item in the correct direction.
3. *Toggle*. Without labels and color status indication, the toggle status is not clear.

It should be noted that 38% of “Design dependence” criterion statements intersected with the “High cost of errors” criterion. For example, P3’s statement about *double slider*: “As long as you can see the values, it is less critical, the user can understand it.”

3. **Existing standards about correct direction**. The criterion means that clear standards or professional knowledge on the correct UI item direction lower the need to evaluate the item’s directionality. This criterion was used mainly to exclude rather than include items in the evaluation set. For example, the participants argued that “mathematical” rules require that numbers, numerical series, or orders should be shown from left to right, and therefore, it is redundant to evaluate UI items containing this kind of information. The direction of such UI items should always be from left to right.

Among the UI items with “mathematic” information were the *trend chart*, *bar chart*, *numerical rating*, and *counter*.

Similar arguments referred to the existence of a “universal convention” by which the designers should be guided. For example, according to P12, there is a global convention that the *vertical scroll bar* should be on the right side: “We were born with the internet scroll bar on the right side, and it has always been there.” Therefore, placing it on the left side would confuse the users, especially when the user is switching between English and Hebrew interfaces: “You browse the sites in English, where there is a scroll bar on the right, then you switch to the site in Hebrew, and suddenly the scroll bar is on the left, and you start looking for it” (P11).

Notably, in 35% of the statements related to this criterion, the experts did not mention a specific standard as a basis for their argument but just claimed that “it is the common” or “obvious” direction. For example, the participants argued that there is no need to evaluate a *calendar including weekdays both in English and BiDi language* since left-to-right is its correct direction: “If weekdays are both in English and Hebrew, it is clear that the calendar direction will be LTR” (P7); “The rule here is so clear, it cannot be otherwise” (P10).

We (the authors) disagree with the rationale provided in some of the experts’ quotes. Our impression is that many of the problems in BiDi interfaces stem from the schism between experts’ and lay users’ knowledge and expectations. Therefore, while we accept the experts’ recommendation about item inclusion according to this criterion, we intend to examine more critically recommendations to exclude items based on it.

4. **Importance of the UI item**. A UI item is important if it is used frequently (e.g., “I use “back” and “next” buttons a lot, these elements were in almost all of my design projects” (P3), or if it is responsible for essential or unique tasks in the interface. In contrast, UI items are less important if they are used infrequently (e.g., “I had never had to deal with *process flow* element in my design” (P4), or if they can be easily replaced by another isofunctional item (e.g., “An icon usually comes with text, so it [the icon] can be easily removed from the interface” (P8). However, the “importance” of a specific UI item may vary by the usage context: “The correct direction of *range elements* is especially important when their purpose is input, and the user need not just read a value but to enter a value. The importance of this task is higher than in the cases where ranges are used only for information display” (P9).

5. **Risk of inconsistent usage**. If a UI item can cause inconsistency which harms the user experience, the directionality of this UI item should be evaluated and defined. Our experts identified three types of inconsistencies that are remarkably similar to those noted by Grudin [10].

Table 2: Top 5 UI items for direction evaluation in BiDi research and for inclusion in the BiDi design system

#	UI item	Graphical item example	Main supporting criteria	Notes and examples of supporting statements
1	“Backward” and “forward” icons (buttons)	 	High cost of errors; Importance of the UI item	Especially, the participants highlighted the frequency of items usage “Almost all interfaces have it” (P3, P9), and their importance for understanding the navigation process: “Incorrect understanding of these icons causes mistakes in system navigation in general” (P4).
2	Date range		High cost of errors	The participants highlighted the ambiguity of the item: “Here we have a cyclical item, so there is no correct start point. If you do not know the direction, you will not understand the information and will make a mistake” (P4, P18).
3	Star rating		Design Dependency	“Without additional indicators, it is a great element for user mistake” (P10).
4	Timeline		Complex UI item Existing Standards	“Because of the combination of numbers and letters, you have no way of knowing the correct and appropriate direction” (P20).
5	Time range		High cost of errors	As in the case of <i>Date range</i> , the participants highlighted the ambiguity and cyclical nature of item: “It is circular, it can be read both from right-to-left and from left-to-right.” (P9), “If not written clearly, there is no way of knowing the correct time range. Most likely you will get it wrong” (P6).

a. Inconsistency with other familiar experiences. For example, “When I was at school, and after that at the university, the charts I saw were depicted from left-to-right. The right-to-left chart does not only confuse me, it is incomprehensible to me, it takes me a lot of time to switch to understanding it” (P14 about *trend chart*).

b. Inconsistency with other UI items within the same interface. For example, the usage of RTL direction in labels (especially texts with ranges) and LTR direction for other elements within the same UI component: “I have a weather app, where the direction of the weekdays is shown right-to-left, and above them, there are additional labels with hours range, and their directionality is left-to-right. . . it is very confusing” (P8 about *calendar with weekdays in Hebrew*).

c. Inconsistency with the same UI item used in LTR language. This occurs when switching between LTR (e.g., English) and BiDi (e.g., Hebrew) interfaces. “The hardest issue is when I receive an invitation to some event from my work calendar, which is left-to-right, and I move to my personal calendar, which uses a different application (right-to-left), and I need to check what is going on this day. Either it takes me a long time, or I’m confused” (P9 about *calendars*).

6. **Problematic UI item in general.** If understanding the information in a UI item takes time in general, regardless of language, its correct direction must be verified and clearly defined to avoid making it even more problematic. The criterion was used as a primary criterion in discussing the *Redo and Undo icons*: “They are confusing

regardless of language. They remind me the Reply and Forward in WhatsApp, you need time to understand them correctly.” (P12).

7. **Complex UI item.** If a single UI item includes multiple visual character strings with opposite directionalities, its correct directionality should be evaluated in the study with BiDi users. An example of such an item could be a *bar chart that includes* “numbers that are written from left-to-right and labels that are written from right-to-left. And then you have a conflict between them which should be checked” (P10).

3.2 Necessary UI items for direction evaluation

The experts’ criteria and recommendations contributed to the prioritization of which UI items should be included in empirical evaluation with BiDi users and in BiDi design systems (a complete list is presented in Appendix). The most recommended items for inclusion are those for which all groups made a unanimous favorable decision. These items appear in Table 2. Five additional items were recommended by 90-95% of the experts to round up the Top 10 list of recommended UI items for empirical evaluation (see Appendix for more detail).

In contrast, the UI items least recommended for inclusion in evaluation studies were icons for *Bus*, *Delivery*, *Phone*, and *Shopping cart*: “Their direction is less important. If we mirror them or not – it does not affect their meaning” (P5, P10). Of the five FGs, only one

recommended that these elements be included in the evaluation. The group members justified their recommendation by noting that the correct icons' directionality could be important in some contexts. For example, actionable icons placed on the screen side (e.g., left) can open a menu only towards the opposite side (right). But it might be misleading if the icon (e.g., a bus) would point to the left.

During the discussions, the participants suggested adding the following items that were not on the original list:

1. *Horizontal scroll bar* – “Verifying its correct directionality is very important. We need to understand how it will be shown with the vertical scroll bar” (P13).
2. *“Search” and “Like” icons* – “Use them a lot in my projects” (P6), “At first glance, I think that there is a universal convention, and the correct directionality is based on the dominant hand. But, maybe in BiDi interfaces, we are discovering another aspect of perception. . . it should be studied” (P15).
3. *Single slider* – “If we should define the direction for the double slider or a numerical rating, it is important to define for it too. Do they match? I’m not sure” (P17).

Finally, we observed a high similarity between the arguments and recommendations of the Russian-speaking group and the Hebrew-speaking groups. For example, the arguments for excluding the *paginator* from the research set: “Usually we see the page numbers, see the direction, and therefore, you can understand, can correct” (Hebrew speaker - P13); “Actually, there is an indication - page numbers, so it’s not so scary, you can correct it” (Russian speaker - P18).

4 CONCLUSION

Five focus groups of HCI professionals from Israel and Russia (N=22) provided seven criteria for future empirical research on the directionality of UI items in BiDi languages. The criteria were used to prioritize which items should be evaluated. The analysis suggests that “*High cost of error*” was the prominent criterion for this prioritization.

The advantage of using multiple experts in evaluating interactive systems and for prioritizing usability issues is well documented [14, 21]. For the unique problem of prioritizing BiDi UI items for empirical evaluation, we chose to discuss the various issues with the experts in focus groups. We observed that listening to arguments by other experts and being involved in group discussions had often caused our participants to reconsider their positions. While previous research stressed the independence of experts' opinions, we believe that, at least in our study's context, the FG method positively affected the quality of opinions and recommendations we received.

The current research is the first in a series of studies. The list of identified items will serve as an input to the second step, in which the desired directionality of the UI items will be evaluated. The findings of this evaluation, in turn, will be the basis for developing design guidelines and design systems for BiDi interactive systems.

This study opens up prospects for further research on the prioritization of design and usability issues. Other studies have proposed various criteria in more general contexts (e.g., [13]). It would be interesting to investigate how a set of general criteria matches criteria that emerged in more specific settings, such as ours. This line

of research could improve both the generalizability of the criteria and their sensitivity to unique contexts.

Our immediate contribution lies in identifying a set of prioritization criteria for evaluating the directionality of UI items in BiDi design systems. In turn, the criteria enabled us to sort the relevant UI items by the urgency with which they should be evaluated. In the longer run, the ensuing evaluation of those items will lead to the development of design systems for BiDi languages to benefit hundreds of millions of BiDi users worldwide.

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A APPENDIX

A1. A list of UI Items sorted by priority

#	UI item	“Include” votes	Number of statements supporting inclusion							Number of statements supporting removal								
			Hi ¹	Dd	Kn	Im	Ri	Pr	Co	Σ	Hi	Dd	Kn	Im	Ri	Pr	Co	Σ
	Backward & Forward* icons	100%	5	-	-	7	-	-	-	9	4	-	-	3	-	-	-	6
	Date range	100%	11	-	6	5	6	2	1	18	-	-	-	-	-	-	-	-
	Star Rating	100%	7	9	-	1	-	6	-	19	6	3	2	-	-	-	-	6
	Timeline	100%	2	-	4	-	3	-	4	9	-	-	-	-	-	-	-	-
	Time range	100%	6	2	3	-	-	-	-	8	-	-	-	-	-	-	-	-
	Calendar without weekdays	95%	7	2	-	-	-	-	-	7	2	-	-	-	-	-	-	2
	1+2 מכצע (“Sale 1+2”)	95%	5	4	3	4	3	2	-	15	-	-	-	-	-	-	-	-
	Bar chart	90%	2	2	-	3	2	-	5	10	3	-	8	-	1	-	2	9
	Numeric Rating	90%	-	-	2	2	3	-	-	5	5	2	5	-	3	-	-	8
	Progress bar*	90%	4	9	-	6	2	-	-	16	2	-	-	7	-	-	-	8
	Back, Next buttons’ order	85%	3	-	-	1	-	-	-	3	-	-	-	-	-	-	-	-
	Redo & Undo icons (buttons)	85%	4	2	-	-	-	7	-	9	3	-	-	-	-	-	-	3
-	Calendar with both BiDi language & English weekdays	85%	6	-	-	2	6	4	-	10	2	-	6	-	-	-	-	8
	Toggle (switcher)	85%	4	8	-	2	2	6	-	13	-	10	-	-	-	-	-	10
	Trend Chart	85%	2	1	-	-	1	-	-	3	-	-	7	1	2	-	-	9
	Calendar with weekdays on BiDi language (e.g., Hebrew)	80%	3	-	-	1	3	2	-	5	1	-	3	-	-	-	-	4
	Process text flow*	80%	1	4	-	-	-	-	-	4	4	5	1	1	-	-	-	7
	Paginator	80%	1	1	-	1	-	-	1	4	4	5	-	2	-	-	-	7
	“Send” icon	80%	4	-	2	2	3	-	-	9	3	1	1	2	-	-	-	5
	Double Slider	75%	1	6	1	-	1	-	-	6	3	3	-	-	-	-	-	4
	2+3 מכצע (Sale 2+3)	75%	4	-	2	1	2	-	-	7	-	2	-	-	-	-	-	2
-	Valid card date” input	75%	3	2	1	-	-	-	-	4	1	1	2	-	-	-	-	3
	Process images’ flow *	70%	1	2	-	-	-	-	1	5	3	3	-	4	-	-	-	5
	Browser scrollbar	60%	3	-	2	1	4	-	-	6	2	-	2	3	2	-	-	5
	Inside-element scrollbar	60%	1	-	1	-	-	-	-	1	-	-	1	1	-	-	-	2
	Minute, price, size, temperature, year ranges	60%	2	2	-	2	-	-	-	5	5	-	2	2	-	-	-	7
	1+3 מכצע (Sale 1+3)	55%	1	-	2	-	2	-	-	3	1	-	3	-	-	-	-	3
	Close button in dialog	45%	2	1	-	-	-	-	-	2	4	1	-	1	2	-	-	5
	Primary, secondary buttons’ order	45%	2	-	-	-	-	-	-	2	2	3	3	-	3	1	-	6
	Carousel *	40%	8	-	1	2	-	-	-	10	7	1	-	2	1	-	-	9
	Navigation drawer and a place of “hamburger” icon*	40%	1	-	-	-	-	-	-	1	2	-	1	4	-	1	-	7
	“Arrival”, “Departure” icons	35%	-	1	1	-	1	2	-	4	5	1	-	1	-	2	-	8
	Counter	25%	2	-	2	-	-	-	-	3	9	-	6	1	-	-	-	14
	“Accessibility” icon	20%	-	-	-	2	-	-	-	2	1	-	-	-	-	-	-	1
	“Bus,” “Delivery,” “Phone,” “Shopping trolley” icons	20%	-	-	-	1	-	-	-	1	4	-	3	4	-	-	-	10

¹ The abbreviations: Hi – “High cost of error”; Dd – “Design dependent UI item”; Kn – “Knowledge about UI item correct direction”; Im – “Importance of the UI item”; Ri – “Risk of inconsistent usage”; Pr – “Problematic UI item in general”; Co – “Complex UI item”; Σ – Total number of statements.

(*) This item was not observed in the list of “problematic UI items” [9] and was included in the current research set as expanding the items categories.